3.3 The Power of Convexity

WORKSHEET

The question

STEP II 2007, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Draw a curve that bends downwards, such as $y = \sin x$ on $(0, \pi)$. Mark two points on it and join them with a straight line. Is the chord above or below the curve?

STAGE 2 Triangles

Ask yourself Which function, which points, and what is their average? Do the angles of a triangle lie in an interval where the function is concave?

STAGE 3 AM–GM in one line

Ask yourself AM–GM compares a product with a sum. Which concave function turns products into sums?

STAGE 4 Using AM–GM

Ask yourself For (a), the left-hand side is a sum of four terms and the right-hand side contains a product of three variables. How many terms should you apply AM–GM to, and why does the 16 help?

Pause and try Now find the minimum of $x^5 + y^5 + z^5 - 5xyz$. (*Hint:* you want a fifth root of $x^5y^5z^5$, so you need five numbers. What should the other two be?)

Try this next

1. **The definition.** (From STEP II, 2018.) Draw a sketch illustrating the chord definition of concavity in the box above. Explain why a function with $f''(x) < 0$ satisfies it. Then, by choosing $t = \frac{1}{3}$, $x_1 = u$ and $x_2 = \frac{v+w}{2}$ suitably, show that a concave function satisfies Jensen's inequality for three points.

2. **A product of sines.** Try part (iii) of STEP II 2018, Question 2 (step.maths.org/questions/18-s2-q2).

3. **Acute triangles.** Show that $\tan x$ is convex on $(0, \frac{\pi}{2})$, and deduce that in an acute-angled triangle, $\tan A + \tan B + \tan C \geq 3\sqrt{3}$.

4 Power means. Show that x^2 is convex, and deduce that $\left(\frac{a_1+\cdots+a_n}{n}\right)^2 \leq \frac{a_1^2+\cdots+a_n^2}{n}$. Interpret this as a statement about the variance of a set of numbers.

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). Chapter 6 is about convexity and Jensen's inequality.
- **Claudi Alsina & Roger Nelsen, *When Less Is More: Visualizing Basic Inequalities*** (Mathematical Association of America). Inequalities proved by pictures, many of them using convexity.
- **Stephen Boyd & Lieven Vandenberghe, *Convex Optimization*** (Cambridge University Press). A modern classic, free online; the first chapters show why convexity matters in practice.

The full guided solution, and what the question leads to at university, is in Section 3.3:
https://beyondstep.keithmcnulty.org/ch03/convexity.html